Gravefin: 5-Week Project Progress Report

Comprehensive Engineering Milestones, Architecture Evolution & UI/GraphQL Modernization

`Timeline: 5 Sprints Completed | Methodology: Adversarial-Safe Architecture | Audit: 100% Passed`

SPRINTS DELIVERED	SYSTEM ROUTES	ESCROW LEDGER	TEST AUDIT MATRIX
5 OF 5 COMPLETED	25 NEXT.JS PAGES	APPEND-ONLY AUDIT	100% GREEN (PASS)

1. Executive Overview & 5-Week Progressive Architecture

Gravefin (Project Graveyard) solves the tragedy of abandoned open-source repositories through an adversarial-safe, escrow-backed marketplace. Over a 5-week engineering trajectory, the platform evolved systematically from foundational database modeling and cryptographic authentication into a complete full-stack web application featuring deterministic state machine automation, bespoke hand-drawn design tokens, Cartesian repository cartography, and a production-grade GraphQL Yoga v5 schema.

5-WEEK ENGINEERING PROGRESSION ROADMAP

WEEK 01 | FOUNDATIONS, DATA MODELING & CRYPTOGRAPHIC AUTH

WEEK 1: 100% COMPLETE

Week 1 focused on formulating the adversarial-safe philosophy, setting up the Next.js 14 Pages Router environment, structuring the relational database schema, implementing anonymous pseudonym authentication, and engineering an archetypal seed dataset.

CORE DELIVERABLES & ACCOMPLISHMENTS:

- `[OK]` **Core Whitepaper & Architecture:** Designed the zero-chat, escrow-backed marketplace model replacing subjective trust with deterministic code deliverables.
- `[OK]` **Relational Schema (Prisma):** Engineered prisma/schema.prisma with 5 models: User, Project, Commitment, Milestone, and LedgerEntry.
- `[OK]` **HMAC Cryptographic Auth:** Implemented lib/auth.js with SHA-256 signed session cookies, bcrypt passwords, and automatic Digger-XXXX aliases.
- `[OK]` **Archetypal Cemetery Seed:** Populated Neon PostgreSQL / SQLite with 8 real-world abandoned project archetypes (Micro-Cache Kernel, Neural Canvas, etc.).
- `[OK]` **Dual Database Flexibility:** Supported production PostgreSQL connection strings and local SQLite dev.db environments seamlessly.

ARCHITECTURAL & INVARIANT HIGHLIGHTS:

* Zero counter-party identity leakage: Diggers and code owners interact strictly through pseudonyms.
* Immutable ledger foundation initialized to record every future credit delta.

PART II: STATE MACHINE AUTOMATION & BESPOKE DESIGN TOKENS

Week 2: Escrow Protocol & The Iron Cron | Week 3: Hand-Drawn Sketch UI System

WEEK 02 | ESCROW PROTOCOL, STATE MACHINE & THE IRON CRON

WEEK 2: 100% COMPLETE

Week 2 established the core deterministic engine of Gravefin. We constructed the 6-state project lifecycle, the append-only financial ledger, automated timeout rules to eliminate ghosting, and a time-travel simulation sandbox for rapid QA validation.

CORE DELIVERABLES & ACCOMPLISHMENTS:

- [OK] **6-State Project Lifecycle:** Defined strict state machine transitions: LISTED -> PENDING_APPROVAL -> ACTIVE -> SUBMITTED -> COMPLETED (plus GHOSTED_RELISTED).
- [OK] **Append-Only Ledger (lib/ledger.js):** Engineered atomic ledger recording STAKE_LOCK, STAKE_REFUND, REWARD_CREDIT, STAKE_FORFEIT, and REP_PENALTY transactions.
- [OK] **The Iron Cron (timeout-check.js):** Engineered deterministic cron automation enforcing 14-day taker delivery deadlines and 7-day owner review timeouts.
- [OK] **14-Day Taker Ghosting Rule:** If taker goes inactive >14 days: collateral forfeited to owner, -20 rep, +1 ghost strike, and project relisted.
- [OK] **7-Day Owner Ghosting Rule:** If owner fails to review >7 days: submission auto-accepted, taker receives full stake refund + completion bonus, owner penalized.
- [OK] **3-Strike Ban Enforcement:** Users accumulating 3 ghost strikes are immediately flagged banned=true and prevented from participating.
- [OK] **Time-Travel Simulator:** Created pages/simulator.jsx console allowing simulated progression (+7d, +14d, +30d) to verify cron execution in real time.

ARCHITECTURAL & INVARIANT HIGHLIGHTS:

* Double-spend prevention: Stake deductions lock balance immediately in escrow before transition.
* Deterministic resolution: No human dispute arbitration required; timeouts govern completion.

WEEK 03 | VISUAL IDENTITY, CUSTOM DESIGN SYSTEM & CORE COMPONENTS

WEEK 3: 100% COMPLETE

Week 3 brought Gravefin to life with a distinctive visual identity. Instead of generic modern tech themes, we designed a warm "Parchment Desk & Hand-Drawn Sketch" design language, complete with custom typography, sketch borders, and interactive protocol components.

CORE DELIVERABLES & ACCOMPLISHMENTS:

- [OK] **Parchment & Sketch Design System:** Configured custom Google Fonts (Patrick Hand, Architects Daughter, Inter), warm parchment backgrounds (#FAF6EE), and ink-line borders.
- [OK] **Tombstone ProjectCard.jsx:** Designed hand-drawn artifact cards featuring revival score badges, stake meters, death autopsy causes, and category pills.
- [OK] **EscrowStatusBanner.jsx:** Built live state-machine progress indicator with visual step indicators, countdown timers, and contextual action triggers.
- [OK] **HandoverChecklist.jsx:** Created dual-party sign-off interface with milestone check-offs and an animated blooming tree visual upon successful verification.
- [OK] **ProvenanceTimeline.jsx:** Implemented cryptographic event chain visually displaying ledger entries, stake locks, and status transitions.
- [OK] **Bespoke SVG Iconography:** Crafted custom hand-drawn SVG icon set in components/Icons.jsx reflecting the archaeology and graveyard aesthetic.
- [OK] **Global Header & Archive Colophon:** Created Navbar.jsx with live credit/rep counters and Footer.jsx carrying the digital graveyard archive philosophy.

ARCHITECTURAL & INVARIANT HIGHLIGHTS:

* Visual clarity: State transitions, locked funds, and countdown deadlines are prominently visible.
* Theme coherence: All buttons, badges, tables, and dialogs adhere to the hand-drawn parchment palette.

PART III: REPOSITORY CARTOGRAPHY, GRAPHQL ENGINE & VERIFICATION

Week 4: Cartography Map & Forensic Diagnostics | Week 5: GraphQL Yoga v5 & Modernized UI

WEEK 04 | DISCOVERY CATALOG, 2D CARTOGRAPHY & FORENSIC AUTOPSY

WEEK 4: 100% COMPLETE

Week 4 delivered the comprehensive user exploration and diagnostic layer. We built the public landing portal, the filterable cemetery catalog, an innovative 2D Cartesian excavation quadrant map, the GitHub forensic intelligence engine, and the specimen workspace.

CORE DELIVERABLES & ACCOMPLISHMENTS:

- [OK] **Public Landing Portal (index.jsx):** Created high-impact hero section, live cemetery telemetry (total buried projects, resurrected codebases, total credits in escrow), and recent listings.
- [OK] **Filterable Catalog (browse.jsx):** Built comprehensive archive search with full-text querying, category tags (AI, Systems, Web3), stake range filtering, and revival score sorting.
- [OK] **2D Cartography Quadrant Map:** Built pages/browse/map.jsx & components/GraveyardMap.jsx plotting specimens across Technical Debt vs. Revival Potential coordinates.
- [OK] **GitHub Forensic Intelligence Engine:** Created lib/githubIntelligence.js calculating Revival Score (0-100) and diagnosing Cause of Decline (Burnout, Scope Creep, Dependency Hell).
- [OK] **Specimen Workspace (project/[id].jsx):** Unified autopsy reports, forensic diagnostic charts, stakeholder action console, and deliverable review in a single workspace.
- [OK] **Dashboard & Workbench Views:** Built pages/dashboard.jsx (owned listings, active claims, transaction history) and pages/workbench.jsx (curatorial marginalia).

ARCHITECTURAL & INVARIANT HIGHLIGHTS:

* Diagnostic transparency: Diggers can review the complete autopsy analysis before locking stake.
* Cartesian discovery: 2D map visualizes high-yield / low-debt revival opportunities at a glance.

WEEK 05 | GRAPHQL YOGA V5 ENGINE, UI MODERNIZATION & QA AUDIT

WEEK 5: 100% COMPLETE (TODAY)

Week 5 finalized the platform with high-performance GraphQL Yoga v5 APIs, major UI modernizations (Git repository URL integration on Submit, Leaderboard removal, and decluttered Navbar with Tools dropdown), and an automated QA test suite that verified 100% of all platform invariants.

CORE DELIVERABLES & ACCOMPLISHMENTS:

- [OK] **GraphQL Yoga v5 Endpoint:** Engineered pages/api/graphql.js with comprehensive typeDefs and resolvers supporting projects, users, commitments, and mutations with GraphiQL IDE.
- [OK] **In-App GraphQL Explorer:** Created pages/graphql.jsx interactive playground with pre-built queries, dynamic JSON response viewer, and latency telemetry.
- [OK] **Submit Page Git Repo Overhaul:** Replaced manual ZIP file upload with primary Git Repository URL and branch inputs. Takers access live repo coordinates upon approved escrow lock.
- [OK] **Leaderboard Removal:** Completely removed pages/leaderboard.jsx and purged all cross-links across Navbar and Footer to eliminate non-essential UI overhead.
- [OK] **De-Cluttered Navigation Bar:** Streamlined Navbar into 4 primary links (Artifacts, Submit, Dashboard, Manifesto) + consolidated floating Tools dropdown with outside-click dismissal.
- [OK] **Comprehensive QA Test Suite:** Executed automated test suite (scripts/test-all.js) and GraphQL tests (scripts/test-graphql.js) achieving 100% pass across all assertions.
- [OK] **Production Build Verification:** Verified npm run build compiles all 25 Next.js static and dynamic routes cleanly with zero syntax or linting errors.

ARCHITECTURAL & INVARIANT HIGHLIGHTS:

* Live code access: Git repository URLs streamline code handover without local file payload overhead.
* Full query flexibility: GraphQL resolvers allow nested querying of projects, authors, takers, and ledger records.

5-WEEK CUMULATIVE DELIVERABLES MATRIX & PROTOCOL CERTIFICATION

Consolidated Engineering Verification | Quality Assurance Audit | Reproduction Coordinates

MASTER 5-WEEK SPRINT SUMMARY MATRIX:

Sprint	Primary Focus Area	Core Artifacts / Key Files	Audit Status
Week 1	**Architecture & Database Modeling**	prisma/schema.prisma, lib/auth.js, prisma/seed.js, CONTEXT.md	VERIFIED (100%)
Week 2	**Escrow State Machine & Cron**	lib/ledger.js, api/cron/timeout-check.js, pages/simulator.jsx	VERIFIED (100%)
Week 3	**Parchment UI & Component Library**	ProjectCard.jsx, EscrowStatusBanner.jsx, HandoverChecklist.jsx, Icons.jsx	VERIFIED (100%)
Week 4	**Cartography Map & Autopsy Engine**	pages/index.jsx, browse.jsx, browse/map.jsx, lib/githubIntelligence.js	VERIFIED (100%)
Week 5	**GraphQL Yoga v5 & UI Modernization**	pages/api/graphql.js, pages/graphql.jsx, pages/submit.jsx, Navbar.jsx	VERIFIED (100%)

CERTIFIED PRODUCTION TECHNOLOGY STACK:

Framework: Next.js 14 Pages Router (25 Routes)

Database & ORM: Prisma 5 + Neon Serverless PostgreSQL / SQLite

API Layer: GraphQL Yoga v5 (/api/graphql) + REST Endpoints

Auth & Security: HMAC SHA-256 Signed Sessions + GitHub OAuth 2.0

Automation: The Iron Cron (Deterministic 14d/7d timeouts)

UI & Styling: Hand-drawn Parchment Design Tokens + Lucide Icons

QA & LOCAL REPRODUCTION VERIFICATION:

[OK]	`npm test`	- Executes scripts/test-all.js (Lifecycle, strikes, escrow locks, OAuth)
[OK]	`npm run test:graphql`	- Executes scripts/test-graphql.js (Yoga v5 schema, queries, mutations)
[OK]	`npm run build`	- Compiles Next.js production bundle across all 25 static & dynamic routes

OFFICIAL AUDIT SIGN-OFF & CERTIFICATION: COMPLETE & VERIFIED

All 5 weekly milestones have been architected, coded, verified, and audited with 100% test pass rates.
Platform is certified for deployment with Neon PostgreSQL, Next.js 14, and GraphQL Yoga v5 endpoints.